

Telmo Barros

Porto, Portugal
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ABOUT

AI and Computer Science researcher with experience in dynamic routing, risk assessment, machine learning and full-stack web development. Currently pursuing a PhD in Computer Science, with applied research experience in projects involving Portuguese Public Administration entities, including inspection planning, route optimization and decision-support systems. Also experienced as a Teaching Assistant in Artificial Intelligence and Machine Learning, combining research, software development and technical communication.

EXPERIENCE

LIACC (FEUP)

Portugal

PhD – Researcher

October 2020 – Present

- Realistic dynamic routing adaptation strategies and custom risk assessment framework.
- Contributed to multiple projects with Portuguese Public Administration entities. ASAE's CIGESCOP, ERS MIA, ERS SAMA4 and EUREKA Retail.

IA.SAE Project – Researcher

October 2019 – September 2020

- Improved the current route generation methodologies performance.
- Worked in a web application prototype directed to ASAE's collaborators.

IA.SAE Project – Masters Dissertation (18 out of 20)

January – July 2019

- Developed the generation of optimal inspection routes module in the IA.SAE project. Implied developing multiple algorithms to solve the optimization problem.
- Helped during the processes of georeferencing and treatment of the provided data.

ONEASIT

Portugal

Full Stack Web Developer

June 2018 – January 2019

- Developed a web portal targeted to the users of the national electricity company of Senegal. From the backend to the frontend layer.
- Integrated the web application with the SAP information system designed for the purpose.

MAISOffice

Portugal

Junior Programmer (Summer Internship)

June 2017 – July 2017

- Helped fixing bugs in the current version of the platform as part of my integration with the software.
- Planned and start developing the 3.0 version from scratch. The old architecture had to be redesigned to integrate an accounting cloud platform alongside the existing CRM and e-business software.

TEACHING EXPERIENCE

University of Porto (Faculty of Engineering)

Portugal

Artificial Intelligence – Teaching Assistant

2022 – 2025

- Taught introductory concepts of Artificial Intelligence and Intelligent Systems to four 3rd-year classes.
- Supervised students in the design and implementation of two introductory projects, focusing on heuristics, adversarial search, supervised and unsupervised learning.
- Contributed to the development of Jupyter notebooks covering AI fundamentals, which were used to support lectures and practical exercises.

- Supported a 1st-year MSc class (~20 students) focused on applying computational learning techniques for decision support.
- Monitored and guided students through the design, implementation, and evaluation of a semester-long supervised learning project.

EDUCATION

Faculty of Sciences of the University of Porto

Portugal

PhD in Computer Science – Artificial Intelligence

October 2020 – Present

Sapienza University of Rome

Italy

Erasmus+ Programme

September 2018 – February 2019

Data Mining, Knowledge Representation and Semantic Technologies, Machine Learning, Web Security and Privacy

Faculty of Engineering of the University of Porto

Portugal

BSc/MSc in Informatics and Computing Engineering (16 out of 20)

September 2014 – July 2019

- Participated in SPE 2016 (Engineering week) exhibiting an academic project to younger students. The project was a game developed in C and Assembly with direct manipulation of the I/O devices.

PUBLICATIONS

Telmo Barros, Alexandra Oliveira, Henrique Lopes Cardoso, Luís Paulo Reis, Cristina Caldeira, João Pedro Machado (2020). “**Generation and Optimization of Inspection Routes for Economic and Food Safety**”, in Proceedings of the 12th International Conference on Agents and Artificial Intelligence - Volume 2: ICAART, Valletta, Malta, February 22-24, 2020, pp. 268–278.

Telmo Barros, Tiago Santos Alexandra Oliveira, Henrique Lopes Cardoso, Luís Paulo Reis, Cristina Caldeira, João Pedro Machado (2020). “**Interactive Inspection Routes Application for Economic and Food Safety**”, in Trends and Innovations in Information Systems and Technologies, pp. 640–649.

Telmo Barros, Alexandra Oliveira, Henrique Lopes Cardoso, Luís Paulo Reis, Cristina Caldeira, João Pedro Machado (2021) “**Economic and Food Safety: Optimized Inspection Routes Generation**”, in: Rocha A.P., Steels L., van den Herik J. (eds) Agents and Artificial Intelligence. ICAART 2020. Lecture Notes in Computer Science, vol 12613. Springer, Cham.

Miguel Milheiro Ferreira, Henrique Lopes Cardoso, Luís Paulo Reis, Telmo Barros, João Pedro Machado (2022). “**Disruption Management of ASAE’s Inspection Routes**”, in Proceedings of the 14th International Conference on Agents and Artificial Intelligence - Volume 3: ICAART, pp. 810–278.

SKILLS

Advanced user of: Java, JavaScript, Python, Flask, Django, SQL, HTML, CSS.

Intermediate user of: C, C++, PHP, Node.js, jQuery, libGDX.

Basic user of: Laravel, Express, Prolog, WebGL.

Languages: Portuguese (native), English (C1).



[Website](#)



[LinkedIn](#)